📄 Internship Project Report

1. **Student Details :**
   Name: Abhishek Kumar
   College / Institution: IGNOU (Patna)
   Course / Branch: BCA
   Internship Program Name: Artificial Intelligence & Machine Learning (AIML) Program
   Project Title: Virtual Paint — Draw in the Air Using Finger Tracking
   Duration of Internship: 45 Days
   Guide / Instructor Name: Shekhar Verma Sir

2. **Project Objective :**

The main objective of this project is to create a virtual painting system where a user can draw in the air using their finger without touching the screen or using a mouse.

This project helps in understanding how computer vision and artificial intelligence can be used to build touchless interactive applications.
The system tracks the movement of the user's index finger and converts that movement into digital drawing on the screen.

3. **Tools and Technologies Used :**

Software

- Python Programming Language
- OpenCV Library
- MediaPipe Library
- NumPy Library
- VS Code Editor

Hardware

- Laptop with Webcam

4. **Methodology :**

In this project, the webcam captures live video continuously.
Each frame from the webcam is processed using the MediaPipe Hands model to detect hand landmarks.

The index finger tip position is used as a cursor to draw on the screen.
A separate virtual canvas is created where drawing strokes are stored so that the drawing does not disappear in the next frame.

Gesture recognition is used to control the drawing process.
When only the index finger is raised, the system goes into drawing mode.
When both index and middle fingers are raised, the system goes into selection mode where the user can select colors or clear the screen from the header panel.

To make the drawing smooth and natural, a deque buffer is used to store previous fingertip positions and draw continuous lines.

5. Implementation :

The project is implemented step by step using Python.
First, the webcam is initialized and frame resolution is increased for better user experience.
Then MediaPipe is used to detect 21 hand landmarks in real time.
Based on the position of these landmarks, finger counting logic is implemented to recognize gestures.
A header user interface is created at the top of the screen which includes different color options, eraser tool, and clear screen button.
When the user touches these options using finger gestures, the drawing tool changes accordingly.
The virtual canvas is merged with the live video frame to display the final output.

6. Results :

The final system successfully allows the user to draw virtually in the air using finger movement.
The system provides:

- Smooth real-time drawing
- Multiple color selection
- Eraser functionality
- Clear screen option
- Gesture-based interaction
- FPS performance display

The drawing experience is interactive and responsive.

7. Learning Outcome :

Through this project, I learned how real-time computer vision systems are developed.
I gained practical knowledge about hand tracking using MediaPipe and image processing using OpenCV.

I also understood how gesture recognition works and how user interaction can be improved using smoothing techniques like deque buffering.

This project improved my debugging skills, logical thinking, and confidence in working with AI-based applications.

8. Challenges Faced :

During the development of this project, I faced several challenges.
Initially, hand detection was not stable and the drawing strokes were not smooth because the fingertip position was changing very fast.

Another major challenge was related to the MediaPipe library.
While implementing hand tracking, I encountered an error where the system showed that **“mp.solutions module was not found.”**
This issue occurred due to Python version compatibility and incorrect library installation.

To solve this problem, I reinstalled the MediaPipe library in a proper virtual environment and used a compatible Python version.
After fixing this issue, hand detection started working correctly.

I also faced difficulty in gesture recognition accuracy and UI interaction handling.
These problems were solved by improving the finger counting logic and implementing stroke smoothing using a deque buffer.

9. Conclusion :

This project shows how artificial intelligence and computer vision can be used to build touchless interactive systems.
The virtual paint system provides an intuitive way of drawing using hand gestures and demonstrates the practical application of AI in real-world scenarios.

The project helped me understand real-time processing, human-computer interaction, and gesture-based control systems.